From the

Transactions of the Silver Society

For the Year C.R. 1101



ON THE TIDES.

By Aldridge, of the Tower of All Things.

Printed for the Silver Society
C.R. 1101

ON THE TIDES.

By Aldridge, of the Tower of All Things.

Communicated to the Silver Society, C.R. 1101.

Severin's Thirteenth Question asks: "Have the risings and fallings of the tides a fixed law?" He asked every city to keep a record of its tides, and to keep it as far as might be in one manner. More than a hundred years later Fanshawe made the mana meter for the Tower of All Things and laid down the rules of the stations, and from that time the cities kept their records in one manner. Those records now fill a hundred and fifty years, and I have spent thirty years in setting them in order. I shall show three things in this paper.

First, besides the ten-year tide and the rise and fall within a day, the risings and fallings of the tide may be resolved into constituent tides of fixed period, of which twelve have periods agreeing with the courses of the twelve mana moons; and there are a few small ones whose frequencies are the sums or differences of those of two moons. Secondly, if it be supposed that the free mana flows as a fluid flows, and that the moons are forces pushing it, a set of equations may be written in the reckoning of flux, from which it follows why the constituents keep fixed periods; and to the terms that ordinary fluids have, one more must be added, which I call the term of self-excitation. Thirdly, the simplified equations, with the term of self-excitation left out, suffice to foretell the ordinary tides from their constituents; but when the crests of several constituents fall together, the foretelling is much in error.

I. THE RECORDS

The stations of the cities are set according to Fanshawe's rules: far from places of casting and from mana pivots; read four times a day, at the fixed hours of midnight, six in the morning, noon and six in the evening; the city, the place and the height of the station recorded; and the records sent to the Tower of All Things. The sensing piece of the mana meter is a fine crystal that resonates with the free mana about it, and the number of divisions by which its needle is turned is what is written down. I have used thirty-seven stations in eleven cities, the earliest begun in C.R. 950, the longest read without a break for a hundred and fifty years. The rest began earlier or later, and some broke off; where a station was moved, its city changed its anchorage or its meter was changed, the old and the new were read side by side for a while according to Fanshawe's rules, and I have joined the two by the difference so found.

The records have their gaps and their errors. I first read together two stations not far apart in the same city; where their difference passed its usual bounds, I searched whether there had been casting or the starting or stopping of a pivot nearby on that day, and struck out what could be so accounted for, keeping and marking what

could not. From this I judge the error of a single reading to be about a third of a division.

Four readings a day suffice to show the rise and fall within the day. Since the four are six hours apart, in their mean the rise and fall that comes once a day just cancels, and this is the mean of the day; in the risings and fallings of the daily means there remain, for the most part, only the constituents whose periods are longer than a day. Where a reading is missing, the mean of that day is left empty, and not made up from three. The constituents spoken of below, except the daily tide, are all found from the daily means.

II. THE CONSTITUENT TIDES

If the risings and fallings of the tide are made up of parts of fixed period, the record of a station may be written

$$h(t) = h_0 + \sum_k A_k \cos\left(2\pi \frac{t}{P_k} - \varphi_k\right) \tag{1}$$

where P_k is the period of the kth constituent, A_k its amplitude, and φ_k its phase. The time t is reckoned in days from midnight at the beginning of the first day of the Month of the Winter Lyre, C.R. 950, and the mean of a day is set at the middle of its four readings, nine in the morning. If the period is known, the amplitude and phase may be found thus: multiply the record by the cosine and the sine of the same period, and sum over the whole record,

$$A_k \cos\varphi_k = \frac{2}{N}\sum_n h(t_n)\cos\left(2\pi\frac{t_n}{P_k}\right), \quad A_k \sin\varphi_k = \frac{2}{N}\sum_n h(t_n)\sin\left(2\pi\frac{t_n}{P_k}\right) \tag{2}$$

N being the number of readings, from which the mean h_0 of the whole record is first taken away. When the period is wrong, the two right-hand sides of (2) tend to nothing as the record lengthens; when it is right, they tend to fixed values. To find the constituents one at a time in this way, the record must be long and their periods well apart; where several lie near together, they must be found jointly, together with the mean. When the period is not known, a series of trial periods is taken one by one, and it is seen at which of them fixed values come out; the longer the record, the narrower the crests, and the closer the trial periods must be set; so I first swept a record of ten years coarsely, to find roughly where the crests lie, and then set the trial periods close about each crest and swept the whole record of a hundred and fifty years. The tables of cosines and sines for the trial periods were given to the star-reckoning arrays of the Tower of All Things to unfold by differences; a table that an array unfolds in a day is worth a month of a company of reckoners. To multiply the readings by the tables and sum them is still the reckoners' work, and every part was reckoned twice, by two men, and compared.

The constituents so found are set out in Table I. The daily tide has the period of a day. The ten-year tide is not a simple cosine: it rises only in the three months of Reawakening, Summer Haze and Scrolls of the tide year, and is almost still at other

times. To make up three months so set apart from cosines of periods ten years, five years, three years and a third, and so on through the whole parts of ten years, would want more than a hundred of them; and the thirtieth and the fifty-fifth parts of ten years are so nearly the periods of the tenth and the eighth moons that not even a hundred and fifty years of records can part them. I have therefore not written it in cosines, but have taken the mean, day by day, of the daily means of those three months in every tide year, save that of 1040, which is mixed with the hundred-year tide, and set it in a table of its own; and in finding the constituents of the moons, the daily means of those three months of the tide years were first set aside. The hundred-year tide came only once in the hundred and fifty years of the records, about 1040; its period I have filled in from the old scrolls and from the chronicles of the lands below, in which its comings are from ninety-eight to a hundred and three years apart, and I take a hundred; its amplitude can be estimated only from that one coming.

Besides these, there are in the daily means twelve constituents whose periods each agree with the course of one mana moon, and Table I gives them in the order of the moons. Their amplitudes are all far smaller than that of the ten-year tide, the greatest less than four divisions and the least a little more than half a division; yet in a hundred and fifty years of records they are all plainly seen. How far the amplitudes may be trusted I judge thus: dividing the hundred and fifty years into five parts of thirty years, I sweep each part by (2). Between the five parts the ten constituents other than those of the twin moons differ by less than a tenth of the amplitude found from the whole record; the two of the twin moons can barely be parted in thirty years, the two crests of the sweep crowding together, and differ between the parts by as much as two tenths. Fig. 1 shows the daily means of the station of Lataki in 1093, with the sum of the twelve constituents and the five combined ones as reckoned from the records of 950 to 1089: the greater part of a year's risings and fallings comes from these constituents.

TABLE I. The constituent tides of the station of Lataki (amplitudes in divisions of the mana meter, phases in degrees)

Constituent	*Period*	*Amplitude*	*Phase*
Daily tide	1 day	14.0	—
Ten-year tide	10 years	rises about 60 in the tide months	—
Hundred-year tide	about 100 years	once, about 1040, rising about 150	—
First mana moon	6.82 days	1.10	40
Second mana moon	11.40 days	2.29	115
Third mana moon	15.97 days	0.90	200
Fourth mana moon (Arcana)	27.32 days	3.60	75
Fifth mana moon	34.85 days	1.40	310
Sixth mana moon	49.10 days	0.69	155
Seventh mana moon (Ankaramu)	27.39 days	2.90	260
Eighth mana moon	66.40 days	1.80	20
Ninth mana moon	88.00 days	0.59	346
Tenth mana moon	121.70 days	2.09	95
Eleventh mana moon	177.30 days	1.19	230
Twelfth mana moon	239.60 days	0.80	180
Fourth and seventh, summed	13.68 days	0.51	334
Second and fourth, summed	8.04 days	0.40	191
Fourth and tenth, summed	22.31 days	0.37	172
Second less fourth	19.56 days	0.39	41
Fourth less tenth	35.23 days	0.36	338

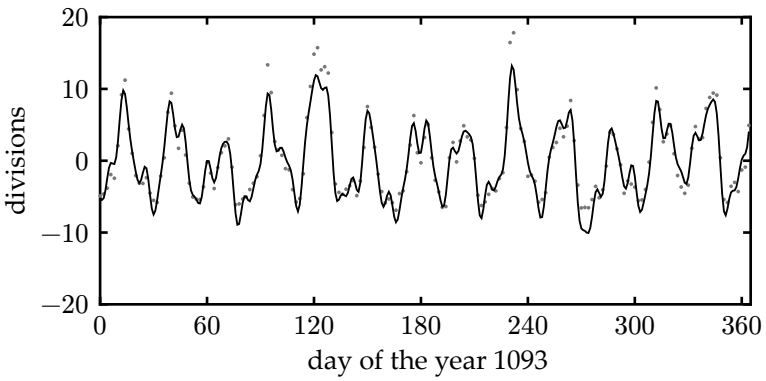


FIG. 1. The daily means of the station of Lataki in C.R. 1093 (points), and the sum of seventeen constituents as reckoned from 950–1089 (line), in divisions of the mana meter

III. THE TWIN MOONS

The fourth mana moon, Arcana, and the seventh, Ankaramu, are twins, and their periods differ by only a fourteenth part of a day. So do the periods of their constituents: taken together, the two first add, then after more than fourteen years all but cancel (their amplitudes being unequal, there remains seven tenths of a division), then after more than fourteen years add again, the whole round taking about twenty-nine years and three months. With records of ten or twenty years, therefore, sweeping by (2) cannot part the two; the reckoning finds only one, and that of an amplitude now great and now small. A record of thirty years can barely part them, and one of a hundred and fifty years parts them plainly. The upper part of Fig. 2 gives the amplitudes found by sweeping (2) over the trial periods for a record of ten years; the lower part enlarges the neighbourhood of twenty-seven days, where the record of ten years shows a single crest, and only the record of a hundred and fifty years parts it into two, one at the period of Arcana and one at that of Ankaramu.

Some before me have said that the tide has nothing to do with the moons, one of their reasons being that the constituent near twenty-seven days is now great and now small, as though it were not driven by a single moon. It now appears that it is the constituents of two moons laid one upon the other.

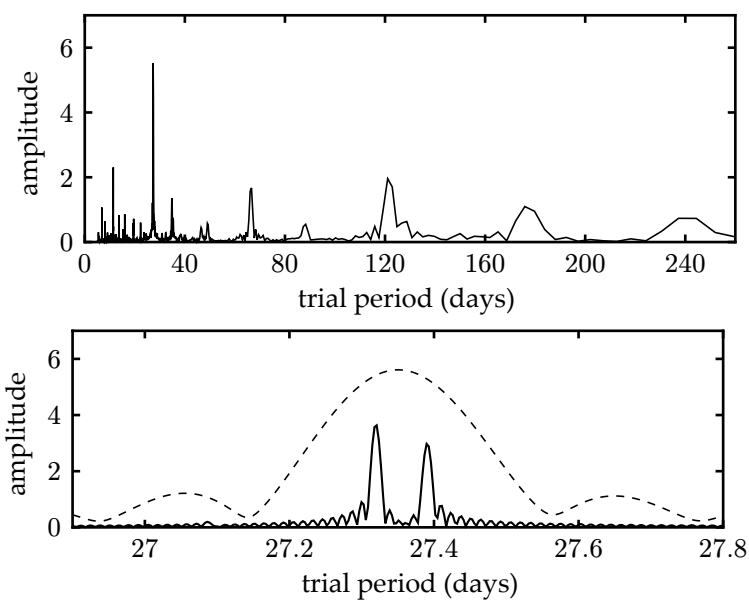


Fig. 2. Above: the amplitudes found by sweeping (2) over the trial periods, for a record of ten years of daily means. Below: the neighbourhood of twenty-seven days enlarged, for the record of ten years (broken line) and of a hundred and fifty years (solid line)

IV. THE COMBINED CONSTITUENTS

The periods of the last five constituents of Table I agree with no moon. The frequency of the constituent of 13.68 days is, within the reckoning, the sum of the frequencies of Arcana and Ankaramu; that of 8.04 days is the sum of those of the second and fourth moons, and that of 22.31 days the sum of those of the fourth and tenth; that of 19.56 days is the difference of the second and fourth, and that of 35.23 days the difference of the fourth and tenth. If each moon pushed only on its own account, the tide should have twelve periods and not their sums and differences. There must therefore be a term that joins two constituents together, and I take it to be in the flow of the mana itself: where both are high, it adds to the tide their product, and the product contains the sum and the difference of their frequencies. The amplitudes of these five go roughly as the products of the amplitudes of the two from which they come, and their phases are roughly the sums or differences of those two phases, which is just the manner of a product. Nor are there only five: other sums and differences of two moons (save the difference of the twins, of which below), and doubles of the frequency of one, are found in the records to the number of twenty and more, all smaller than these five, the greatest about a third of a division; I have set only the five greatest in Table I, and the rest in a table kept at the Tower of All Things. Every combined constituent in which Arcana has a part has beside it one in which Ankaramu takes its place, differing in frequency only by the difference of the twins; and these, like the twins, are plainly parted only in a hundred and fifty years of records. The difference of the twins has the period of twenty-nine years and three months spoken of above; there are only five rounds of it in a hundred and fifty years, and it is mixed with the rising and falling of the one hundred-year tide about 1040, so that I cannot part it out.

V. THE TIDE EQUATIONS

I suppose that the free mana flows in the world as a fluid flows, from the denser places to the thinner, and that the twelve mana moons are forces pushing it, each with its own period. On this supposition two equations may be written in the reckoning of flux. Let ρ be the density of the free mana at a place, and $\boldsymbol{v}$ the velocity of its flow. Mana neither comes from nothing nor goes to nothing, which is Finch's general account; therefore

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{v}) = 0 \tag{3}$$

The flow of the mana is governed by four things: the push from denser to thinner, whose strength is as the slope of the density; the drag of the flow upon itself, which slowly evens out places of unequal speed; the pushing of the moons; and the term of self-excitation:

$$\frac{\partial \boldsymbol{v}}{\partial t} + (\boldsymbol{v} \cdot \nabla)\boldsymbol{v} = -a\nabla\rho + \nu\nabla^2\boldsymbol{v} + \sum_{k} \boldsymbol{F}_k + \kappa\rho\nabla\rho \tag{4}$$

where a, ν and κ are three positive constants, and $\boldsymbol{F}_k$ is the push of the kth moon, rising and falling with its period as a single cosine. The first three terms ordinary fluids have also; the last, $\kappa\rho\nabla\rho$, is the one I add. It says that mana has a disposition to gather towards where it is denser, and gathers the more strongly the denser it is. It does not add mana from nothing, since (3) holds as before: where the mana grows more, it has flowed in from round about; the term only carries mana towards the denser places.

The term of self-excitation runs against the first term. Taken together the two are $-(a-\kappa\rho)\nabla\rho$: while the density is not great, the push is still from denser to thinner; when the density passes $\frac{a}{\kappa}$, the push is turned about, and the denser place gathers ever denser. I take a local mana surge to be a stretch of air in which the density has passed this bound. When the ten-year and the hundred-year tides rise, the density over a wide stretch is raised together and comes nearer this bound, yet in the records no wide stretch has passed it; and it is perhaps for this reason that the local surges come mostly in the tide years.

I add this term for two reasons. The first is the combined constituents of Section IV. Ordinary fluids also have two terms that can join two constituents together, the term $(\boldsymbol{v}\cdot\nabla)\boldsymbol{v}$ of (4), and in (3) the product of the rising and falling of the density with the velocity; but the tide flows so slowly, and rises and falls so little, that, by the speeds and risings I have estimated, the combined constituents these two give are less than a tenth of those found. With the term of self-excitation it is otherwise. Let ρ_0 be the usual density, and ρ' its rising and falling about it. Taken with the first term, the term of self-excitation gives $-(a-\kappa\rho)\nabla\rho$, in which the part that goes as the square of ρ' stands to the part that goes as ρ' in the ratio $\frac{\kappa\rho'}{a-\kappa\rho_0}$. The nearer the usual density comes to $\frac{a}{\kappa}$, the smaller the divisor and the greater the ratio; and only once the usual density is past the half of $\frac{a}{\kappa}$ are the combined constituents that the term of self-excitation gives greater than those of the two terms of ordinary fluids. Those found are more than ten times what these two give, so that to make them up the usual density must lie much nearer still to $\frac{a}{\kappa}$. The second is three unusual risings in the records, at Varen in the Month of the Winter Lyre of 1012, at Onais in the Month of Summer Haze of 1047, and at Lataki in the Month of Reawakening of 1080, when the crest stood more than four tenths higher than the constituents, as reckoned from the tables, could then have raised it. Each of the three was recorded at a single station only, and I do not presume to say that none was misread; I say only that they agree with the term of self-excitation.

VI. THE SIMPLIFIED EQUATIONS, AND FORETELLING

Put $\rho=\rho_0+\rho'$, taking ρ_0 to be the same everywhere and unchanging. When the term of self-excitation is taken with the first, the part that goes as ρ' has the same form as the first term, only with $a-\kappa\rho_0$ in place of a, which I write b; the part that goes as the square of ρ', since in ordinary tides ρ' is far less than $\frac{a}{\kappa}-\rho_0$, is far smaller than the first term, and may be left out together with the two terms of ordinary fluids. With ρ' and $\boldsymbol{v}$ small, equations (3) and (4) become

$$\frac{\partial \rho'}{\partial t} + \rho_0 \nabla \cdot \boldsymbol{v} = 0, \quad \frac{\partial \boldsymbol{v}}{\partial t} = -b\nabla\rho' + \nu\nabla^2\boldsymbol{v} + \sum_k \boldsymbol{F}_k \tag{5}$$

These are the simplified equations. Each of their terms, but the push of the moons, is proportional to ρ' or $\boldsymbol{v}$, so that the push of each moon, once it has been long at work, raises at each place only a rising and falling of the same period, with an amplitude and phase that differ from place to place: and this is why the constituents keep fixed periods. This may be seen in the records of the several stations. Table III gives the constituent of Arcana at four stations: its period, swept out at each station by itself, is everywhere 27.32 days to within a hundredth of a day, but its amplitude and phase differ from station to station. The push is that of one and the same moon, and the answer differs from place to place, as (5) says; were the risings and fallings of the tide only the confusion of each place by itself, the period would not be everywhere the same.

TABLE III. The constituent of Arcana at four stations (period everywhere 27.32 days)

Station	*City*	*Amplitude*	*Phase*
Tower Top	Lataki	3.60	75
East Harbour	Kanaich	2.85	102
North Dyke	Onais	4.10	61
Outer Ring	Varen	3.25	118

The amplitudes and phases at each place cannot yet be reckoned exactly from the constants of the equations, and must be found from the records of each station by (2); once found, the tide may be carried forward by (1). For daily foretelling the equations need not be solved: it suffices to carry forward the tables of constituents of each station by (1), adding the ten-year and the hundred-year tides from their own tables; what the simplified equations tell is why such carrying forward may be trusted.

VII. THE TRIAL

From the records of 950 to 1089 I found the constituents of every station, and then reckoned by (1) the years 1090 to 1099, and compared the reckoning with what was measured in those ten years. For this trial the number of the constituents, their periods and the manner of striking out readings were all settled from the records before 1090, and those after were not looked at. The result for the station of Lataki is given in Table II (the three tide months of 1090 are spoken of below, and the days whose means are empty are not counted); those of the other thirty-six stations are much the same, the greatest differences all falling on the days when the crests were piled. On ordinary days the reckoning and the measure differ on the mean by less than a division; when the crests of two or three constituents come near together, they differ by two divisions; and in the thirty days when the crests of several constituents fell together in one place, by nearly seven divisions, and on the worst day by more than thirteen. The higher the crests were piled, the lower the

reckoning fell. This is just the manner of the term of self-excitation: the simplified equations leave it out, which does no harm in ordinary times, but cannot hold the tide down when it is piled high.

Table II. The constituents of 950–1089 carried forward to the daily means of the station of Lataki in 1090–1099: differences from the measure, taken without sign, in divisions

Case	*Days*	*Mean difference*	*Greatest difference*
Ordinary	2,487	0.7	3.3
Crests of two or three constituents near together	1,004	2.0	7.3
Crests of several constituents fallen together	30	6.8	13.4

1090 was a tide year, and in its three months the reckoning of the ten-year tide also fell somewhat low; in the tide year 1100, which lies outside the ten years of the trial, it fell lower still, and this I leave to be checked by the records to come.

VIII. WHAT REMAINS UNSETTLED

First, whether the moons are the cause of the tides. This paper writes the moons as forces pushing the mana, because, so written, the tides can be reckoned. Others hold that the moons and the tides are alike shadows of the Sea of Mana, and that the waxing and waning of the moons are only signs of the tides. On that view the forces in the equations should be replaced by something else; but the constituents found would not change, so that from the records of the tides alone the two views cannot be told apart.

Secondly, the term of self-excitation. Its grounds are the combined constituents and the three unusual risings, and both are still slight. The constant κ, and how far the usual density lies below $\frac{a}{\kappa}$, I can estimate only roughly, from the amplitudes of the combined constituents. If in the records to come the unusual risings do not appear again, or if the combined constituents have another cause, the term should be struck out.

Thirdly, the hundred-year tide. It came only once in the records, and both its period and its amplitude are filled in.

Fourthly, the local mana surges. The equations say that when the density passes $\frac{a}{\kappa}$ it gathers ever denser; but in the moment before it passes, the records show no sign, so that, with the records and reckonings we have, warning can be given only after the surge has begun, and it cannot yet be foretold.

Fifthly, the solutions of the equations. With the term of self-excitation the denser places gather ever denser, and whether a solution may cease within a finite time to be smooth, and bring forth an infinite density, I do not know. I know only that in the records of ordinary tides no loss of smoothness is to be seen.

IX. THE TIDE-DISC

To reckon a year of tides by (1), seventeen cosines must be added up day by day, which is hard labour for the reckoners. A tide-disc may be made: seventeen water-

wheels, joined by gears whose teeth are so counted that each turns with the period of one constituent, and each driving a piston up and down in a cylinder of water, the throw of each set by its amplitude and its starting point by its phase; the seventeen cylinders open below into one narrow pipe, so that the water each drives in and draws out is gathered there, and the height of the water in the pipe is the sum of the seventeen risings and fallings, which a float draws as the height of the tide. The ten-year and the hundred-year tides are added afterwards from their own tables. I have asked the craftsmen of the Tower of All Things to make trial of such an instrument.